

EN401 — Handout 1: Energy Modelling Concepts

Reference Energy System (RES) Architecture & Technical Units

1. Conceptual Overview & System Scope

Handout 1 introduces the fundamental structure of energy systems linear programming. It defines the energy chain linking primary resource extraction through transformation technologies to final useful demand.

2. Standardized Units & Conversion Ratios

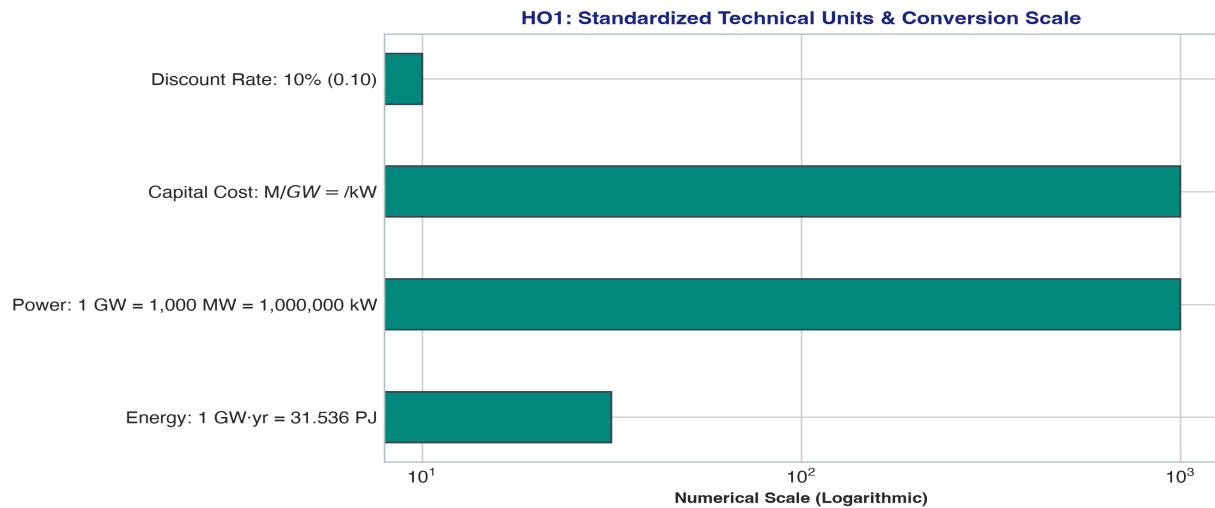


Figure: Technical units standardization framework in OSeMOSYS.

3. Core Takeaways

- Established standard units: Petajoules (PJ) for energy and Gigawatts (GW) for power.
- Formulated the qualitative Reference Energy System linking resources to consumer demand.